Based on the tolerance to intensities of light, the plants are divided into two types. They are

1. **Heliophytes** - Light loving plants. Example: Angiosperms.
2. **Sciophytes** - Shade loving plants. Example: Bryophytes and Pteridophytes.

In deep sea (>500m), the environment is dark and its inhabitants are not aware of the existence of celestial source of energy called Sun. What, then is their source of energy?



Palaeoclimatology–Helps to reconstruct past climates of our planet and flora, fauna and ecosystem in which they lived. Example: Air bubbles trapped in ice for tens of thousands of years with fossilized pollen, coral, plant and animal debris.

b. Temperature

Temperature is one of the important factors which affect almost all the metabolic activities of an organism. Every physiological process in an organism requires an optimum temperature at which it shows the maximum metabolic rate. Three limits of temperature can be recognized for any organism. They are

1. **Minimum temperature** - Physiological activities are lowest.
2. **Optimum temperature** - Physiological activities are maximum.
3. **Maximum temperature** - Physiological activities will stop.

Based on the temperature prevailing in an area, **Raunkiaer** classified the world's vegetation into the following four types. They are megatherms, mesotherms, microtherms and hekistotherms. In thermal springs and deep sea hydrothermal vents the average temperature exceed 100°c.

Based on the range of **thermal tolerance**, organisms are divided into two types.

1. **Eurythermal:** Organisms which can tolerate a wide range of temperature fluctuations.

Example: *Zostera* (A marine Angiosperm) and *Artemisia tridentata*.

2. **Stenothermal:** Organisms which can tolerate only small range of temperature variations. Example: Mango and Palm (Terrestrial Angiosperms).

Mango plant does not grow in temperate countries like Canada and Germany.

Thermal Stratification

It is usually found in aquatic habitat. The change in the temperature profile with increasing depth in a water body is called **thermal stratification**. There are three levels of thermal stratifications.

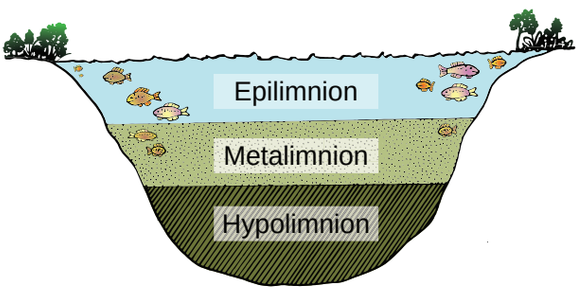


Figure 6.3: Thermal stratification of pond

1. **Epilimnion** – The upper layer of warmer water.
2. **Metalimnion** – The middle layer with a zone of gradual decrease in temperature.
3. **Hypolimnion** - The bottom layer of colder water.

Temperature based zonation

Variations in **latitude** and **altitude** do affect the temperature and the vegetation on the earth surface. The latitudinal and altitudinal zonation of vegetation is illustrated below:

Latitude: Latitude is an angle which ranges from 0^0 at the equator to 90^0 at the poles.

Altitude: How high a place is located above the sea level is called the altitude of the place.

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